

Fernando Jr. Villanueva

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<https://fvjr156.github.io/my-portfolio>

Passionate software developer with freelancing experience in crafting high-performance mobile applications using Java, Android Studio, and Room Databases, alongside scalable web applications powered by React, Express, MySQL, and Node.js.

Skills

- Backend Web Development
- Frontend Web Development
- Mobile App Development
- Research and Technical Documentation
- MySQL, MongoDB, SQL Server, SQLite, Room
- Git and GitHub
- Tailwind CSS and Bootstrap
- HTML and CSS Proficiency
- JavaScript and TypeScript Proficiency
- Java and C# Proficiency
- C/C++ and Low-Level Programming Familiarity
- MERN stack and MVC architecture familiarity

Projects

BridgeXSign — ML-Based Sign Language Translator Application

May to December 2025

- Engineered an Android application that translates American Sign Language gestures into text and speech, leveraging a custom TFLite model for high-speed, gesture inference for edge devices.
- Optimized on-device gesture recognition using a MediaPipe Hands pipeline, ensuring low-latency camera-to-data conversion, and a smooth, responsive user experience.
- Integrated text-to-speech and accessibility features, allowing for seamless multi-modal communication between sign language users and non-signers.
- Designed an intuitive, high-performance user interface designed for camera-and-model-intensive workflows, ensuring fast, seamless, and responsive functionality even on limited mobile hardware.
- Managed the end-to-end ML lifecycle, from optimizing the TFLite for mobile deployment, to handling real-time coordinate mapping of hand landmarks for improved gesture accuracy.

ARISE — Augmented Reality PC Assembly and Disassembly Application

May to August 2025

- Developed an immersive AR Android application designed to simulate complex PC assembly and disassembly procedures through interactive 3D visualization.
- Built an AR environment using Unity, and implemented high-precision marker recognition using Vuforia, to trigger 3D model spawning and real-time rendering.
- Engineered an interactive component system that allows users to observe, scale and rotate PC parts (CPU, GPU, RAM, etc.), simulating realistic spatial placement and assembly sequences.
- Optimized 3D asset performance for mobile hardware, ensuring smooth rendering and responsive model tracking during component interactions

- Designed a guided educational workflow that bridges the gap between digital simulation and physical hardware handling, providing a safe and cost-effective training environment for tech students.

ARCMS — Automated Response Collection and Monitoring System for Cucina de Marquina

January to June 2024

- Built a full-stack web application that automates data ingestion from Microsoft Forms via a Power Automate pipeline, streamlining exports of raw form responses data into a centralized MySQL database.
- Developed a high-performance dashboard using React and Material UI, with real time data visualization with Chart.js for immediate and easy monitoring of incoming data.
- Built an Express.js backend to handle serialized JSON data, implementing RESTful API endpoints for efficient data insertions and queries.
- Accelerated reporting workflows by designing granular analytics modules (which are categorized by form, individual response, and question), reducing manual data synthesis time and providing deeper insights than native Microsoft platform tools.

Vremia — Digital Time Management Application

March to June 2023

- Developed a localized time-management Android application focused on scheduling and task planning with fully-offline functionalities.
- Built a persistent data layer using SQLite, implementing local database management to ensure zero-latency offline data access.
- Designed a responsive UI following Material Design and HCI principles, optimizing touch targets, icon density, and adaptive layouts for a consistent user experience across different devices.
- Leveraged the Android Notification Service to deliver priority-level alerts, ensuring critical reminders are successfully triggered and visible to the user in real-time.
- Integrated Material Component Library for intuitive, consistent styling across form components and dialogs such as date and time pickers, to minimize user errors and streamline the scheduling process.

Word Experience Data Encoder Intern — Student-Trainee

National Bureau of Investigation, Marikina City Satellite Office
February to March 2024 (1 month)

- Requirement for SHS Work Immersion course
- Responsibilities:
 - Processed data for 500+ NBI clearance applicants (up to 50 applicants per day), ensuring 100% accuracy in record digitization.
 - Verified identities of 500+ applicants through submitted legal document cross-checks (e.g. birth certificates and past clearance applications), upholding data entry accuracy.
 - Guided 300+ applicants through clearance application processes, streamlining application submissions and resolving queries to improve turnaround times.
 - Contributed to a positive work environment by proactively assisting colleagues as needed, fostering strong professional relationships and a collaborative atmosphere.
 - Developed strong attention-to-detail skills through the constant review of entered information, ensuring high-quality results.
 - Used computer software to encode, store, and retrieve data.
 - Maintained strict and utmost confidentiality while handling sensitive client information according to company guidelines and legal regulations.

Education**Bachelor of Science in Computer Science**

STI College Marikina
Expected finish March 2028

Years 1-2:

- Developed multi-platform applications using ASP.NET (web/mobile/desktop), Android Studio (Java), and Java AWT, targeting diverse devices and operating systems.
- Managed SQL Server databases using SQL Server and SQL Server Management Studio, with advanced queries, stored procedures, and database normalization.
- Implemented data structures (e.g. arrays, trees, graphs) and algorithms. Designed and analyzed efficient algorithms for real-world scenarios.
- Applied Object-Oriented Programming (OOP), Functional Programming, and MVC architecture in coding exercises and hands-on projects, enhancing code modularity, scalability, and maintainability.

SHS — IT in Mobile App and Web Development Strand

STI College Marikina

August 2022 – July 2024

- Mastered 3 core programming languages (Java, C#, JavaScript) through hands-on projects and lessons, building foundational coding proficiency.
- Developed interactive web applications using the web technologies HTML, CSS, and JavaScript, implementing responsive web designs and smooth client-side functionality.
- Applied Object-Oriented Programming (OOP) and Functional programming concepts in hands-on projects, lessons, and coding exercises, optimizing code structure, modularity, scalability, and maintainability.
- Built one (1) mobile application using Android Studio and Java, and one (1) full-stack web-based application using MERN stack and MVC architecture, as parts of major course deliverables.
- Designed and queried relational databases using SQL Server and SQL Server Management Studio, executing CRUD operations and complex joins for data-driven projects.

Certifications

- **Crash Course on Python**
issued by Coursera — April 2026
<https://coursera.org/share/4d166512610af4021eee04ae9c640856>
- **Foundations of Agile Project Management**
issued by HackerRank — March 2026
<https://coursera.org/share/2592eec1c05cf07d832f044d8ff2de6f>
- **React (Basic)**
issued by HackerRank — November 2024
<https://www.hackerrank.com/certificates/6e931408ac0a>
- **Java (Basic)**
issued by HackerRank — November 2024
<https://www.hackerrank.com/certificates/19689f072134>

Languages

- **English** – Fluent, Working Proficiency
- **Filipino (Tagalog)** – Native